Field fix procedure front lower arm inspection



Part name	Part No.
Arm, Assy lower (LH)	4013A335
Arm, ASSY lower (RH)	4013A336

Special Tools

Package of 5 pieces



Gap & width gauge



Welding line position gauges

LF



RF



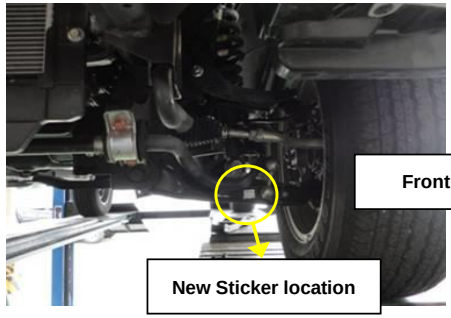
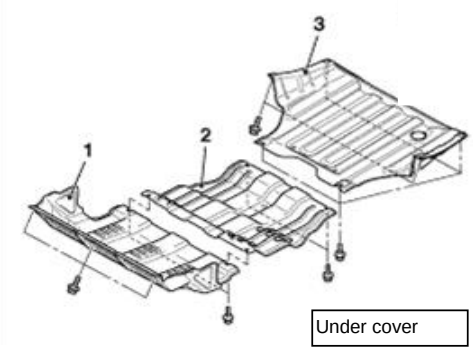
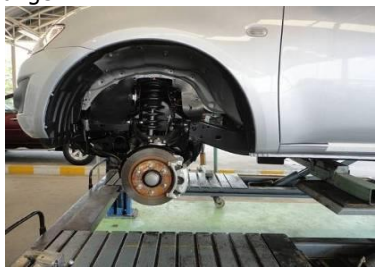
LR

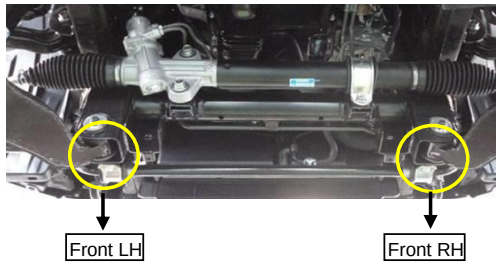
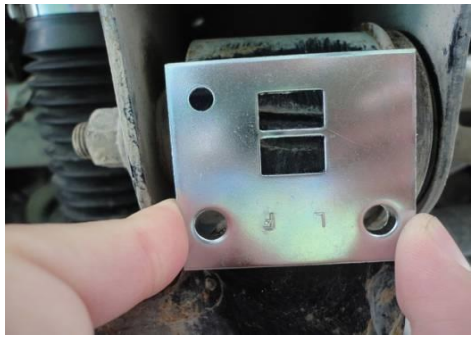
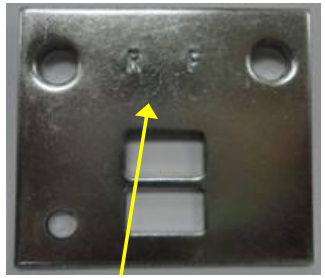
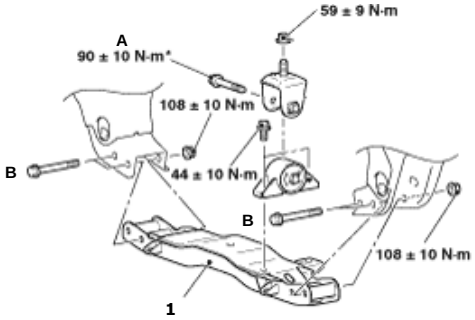


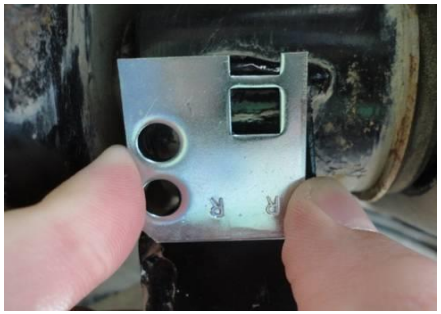
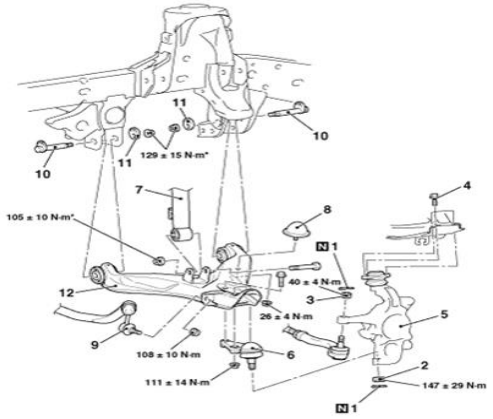
RR



No.	SUBJECT / PICTURE	METHOD	KEY POINT	TOOL
	Inspection lower arm welding line method			
1	**Lift up vehicle**  1.1 Multi-post type lift  1.2 Lifter type: two-post type lift (not support on wheels)	**Removal steps** 1. Lift up vehicle		

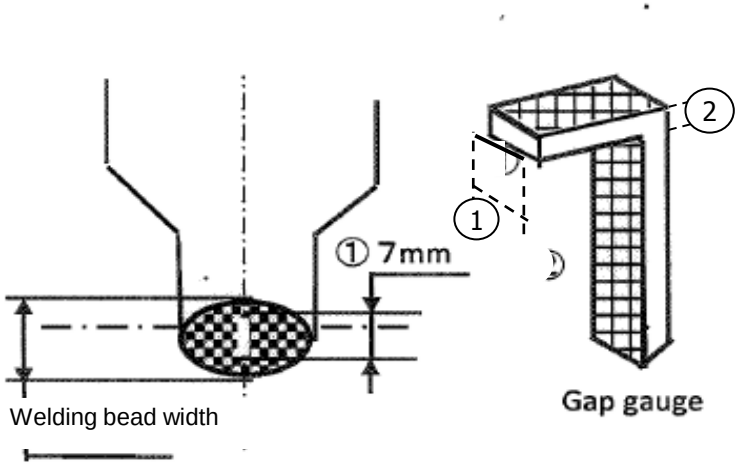
No.	SUBJECT / PICTURE	METHOD	KEY POINT	TOOL
2	**Check paint mark or sticker**   New Sticker  Old Sticker	Check step 1. Check if there is any paint mark on welding bead of inspected locations. If paint mark is present, inspection by gauge is NOT necessary. If paint mark is absent, go to "2. Check by Sticker." 2. Check by Sticker If lower arm stickers are the new type, inspection by gauges is NOT necessary. If lower arm has neither paint mark nor new sticker, go to step.3  	- Lower arm with paint mark is produced after the countermeasure. - Lower arm with new sticker is produced after the countermeasure. LH Rear side LH Front side  Rear side of vehicle Front side of vehicle	
3	**Remove wheel assy and undercover**  Under cover	**Removal steps** 1 In case of working on a **two-post type lift**, remove front wheels. 2. Remove "Under skid plate" (1) 3. Remove "Engine room under cover front" (2) 4. Remove "Engine room under cover rear" (3) (No.4 process applies to models with "Engine room under cover rear")	- Remove wheel assy on two-post type lift for the prevention of wheel alignment change. 	

No.	SUBJECT / PICTURE	METHOD	KEY POINT	TOOL
4	**Inspect lower arm welding bead (Front side)**  	**Inspection step** 1. Check welding bead on Front RH side (1) Use "Gap and width gauge ① to check welding bead width at the center of welding line. Welding bead width is OK if it is wider than the gauge (≥ 7mm). Please refer to judgment criteria **(Attached Sheet 1)** (2) Use Gap and width gauge ② to check gap between lower plate and pipe. Welding bead is OK if gauge cannot reach it. Please refer to judgment criteria (**Attached Sheet 2**.) (3). Check welding bead position of Front RH side with Welding line position gauge "R F". If welding bead is seen in both gauge windows, it is OK. Please refer to judgment criteria (**Attached Sheet 3**) 2. Check welding line of Front LH side Check welding line of Front LH side in the same manner as RH side. * If both RH and LH side welding lines are NG, replace both lower arms with new parts (don't remove front suspension crossmember). Please go to step no.7	**Caution** When inspecting welding position, select a designated gauge for each position  R F = Front right side. R R = Rear right side. L F = Front left side. L R = Rear left side. In case welding line is judged NG, replace lower arm with a new one.	Inspection gauge
5	**Remove front suspension crossmember**  	1. Remove bolt (A) in case of 4WD model. 2. Remove bolts (B) 4 pieces. **3. Remove front suspension crossmember (1)**		

No.	SUBJECT / PICTURE	METHOD	KEY POINT	TOOL
6	**Inspect lower arm welding line (Rear side)**  	**Inspection step** 1. Check welding line Rear RH side (1) Use "Gap and width gauge ① to check welding bead width at the center of welding line. Welding bead width is OK if it is wider than the gauge (≥ 7mm), . Please refer to judgment criteria **(Attached Sheet 1)**. (2) Use Gap and width gauge ② to check gap between lower plate and pipe. Welding bead is OK if gauge cannot reach it. Please refer to judgment criteria **(Attached Sheet 2)**. (3) Check welding bead position of Rear RH side with Welding line position gauge "R R". If welding bead is seen in both gauge windows, it is OK. Please refer to judgment criteria **(Attached Sheet 3.)**. 2. Check welding line of Rear LH side Check welding line of Rear LH side in the same manner as RH side. * If welding bead is OK on both sides (RH and LH), re-install front suspension crossmember, under cover and wheels according to the workshop manual.	**Caution** When inspecting welding position, select a designated gauge.  R F = Front right side. R R = Rear right side. L F = Front left side. L R = Rear left side. In case welding line is judged NG, replace lower arm with a new one.	Inspection gauge
7	**Replace front lower arm**  	In case welding line is judged NG, replace lower arm with a new part 1. Replace lower arm with a new one in accordance with the procedure shown in workshop manual. 2. Install front suspension crossmember, under cover and wheels 3. Check wheel alignment and make necessary adjustment.		

Inspection gauge judgment criteria

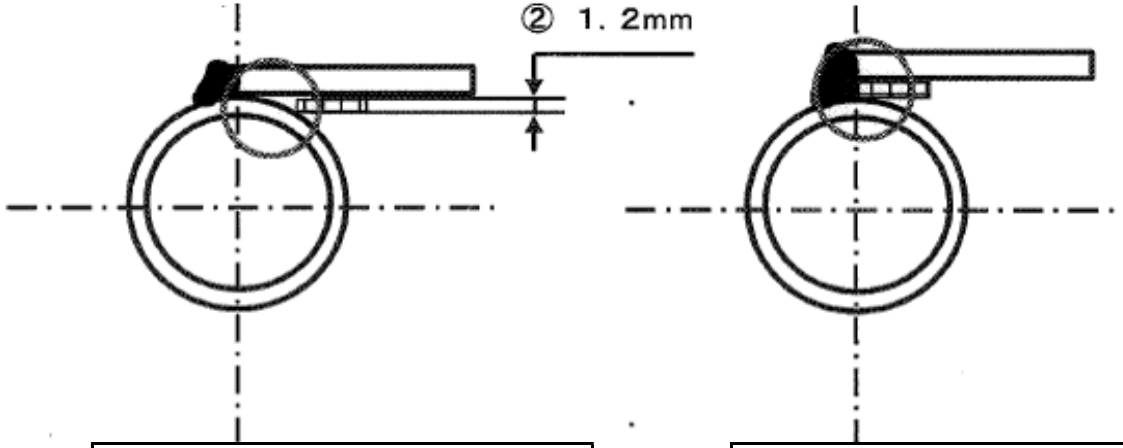
1. Inspection of welding bead width by gap & width gauge (1)



Judgment Criterion
Apply ① section of Gap Gauge at the center of welding bead. The bead width must be wider than the gauge.
OK

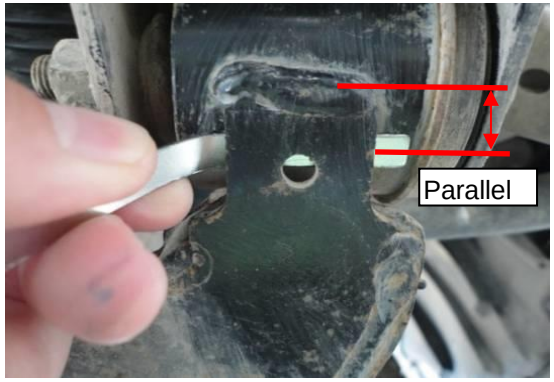


2. Inspection of welding gap between lower plate and pipe by gap & width gauge (2)



Judgment Criterion
Gauge doesn't reach welding bead.
OK

Judgment Criterion
Gauge touches welding bead.
NG



* Note
- Insert guage paralell with welding bead.
- Use shorter side of gauge because longer side of gauge can not be inserted into rear side inspection point.

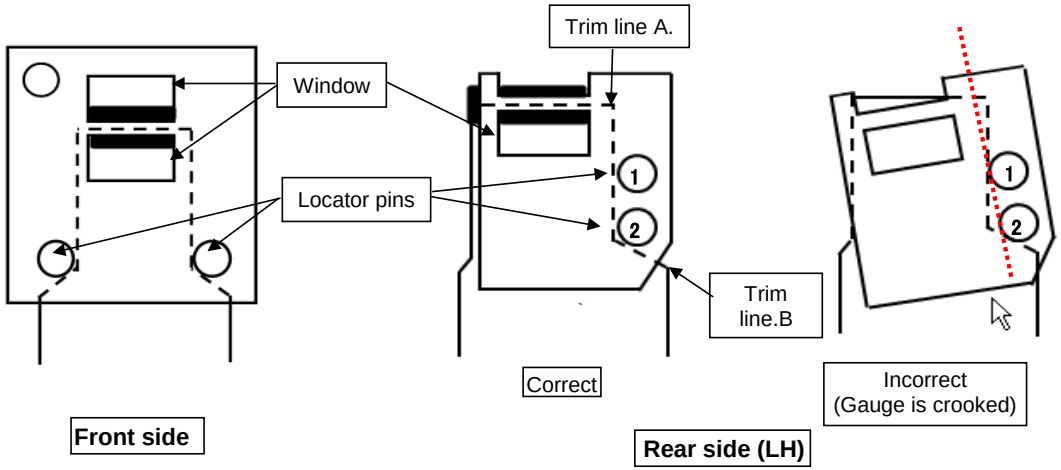


Shorter side

3. Inspection of welding line position by welding position gauge

Set gauge as shown below and see welding bead through the windows

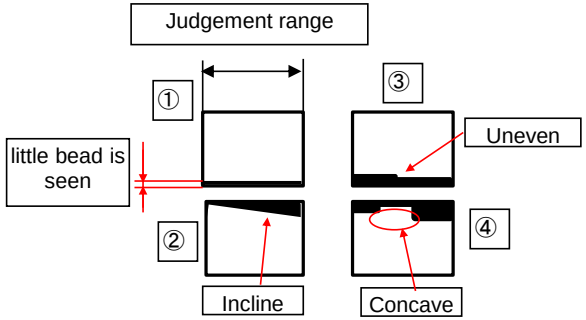
(1)Set gauges on lower



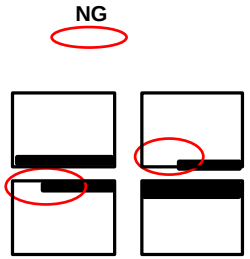
*Key points
Make sure locator pin #1 is touching trim line at A.
Make sure locator pin #2 is touching trim lines at both A and B.



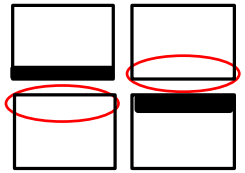
(2)Check welding bead through the windows.



Judgment Criterion	
Welding beads is seen through both windows for the entire length.	OK



Judgment Criterion	
Welding bead is seen partially	NG



Judgment Criterion	
Welding beads is seen through only one of windows.	NG

***Key points in checking the welding bead.**

As long as bead is seen for the entire length of the window, it should be judged OK even when the following conditions are present.

① The amount of welding bead shown through the window is little
② Welding bead is inclined.
③ Welding bead shape is uneven.
④ There is a concave in welding bead.